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Enhancing Classroom Attendance Systems with Face Recognition through CCTV using Deep Learning

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Abstract

This paper investigates the effectiveness of face recognition-based attendance systems, focusing on deep learning models applied to automate attendance tracking in classroom settings. Attendance management is critical for educational institutions, but traditional methods are often plagued by inefficiencies and inaccuracies. To address these issues, we explored the use of advanced face recognition technology, particularly leveraging CCTV-based systems combined with deep learning models namely ResNet-50 and VGG-16. We used images collected from actual classroom environments, applied image dataset augmentation methods, used transfer learning for features extraction and Principal Component Analysis (PCA) for dimensionality reduction and then used the data as an input in the fine-tuned pre-trained models. Our experimental analysis using PCA as a feature extractor indicate that the ResNet-50 model achieved precision and recall rate of 96% and 93% respectively and VGG16 achieved precision and recall rate of 92% and 89% respectively. Despite promising results, challenges remain, such as the impact of camera quality and inherent limitations such as viewing geometry in current face recognition technologies. Future improvements may involve upgrading camera systems and employing image enhancement techniques. This study offers insights into the practical application of face recognition for attendance and suggests directions for enhancing system reliability and accuracy.

Keywords: Automated attendance, face recognition, PCA, Deep Learning, CCTV, VGG-16, RESNET-50

1. Introduction

Attendance refers to the practice of recording and tracking the presence of individuals within an organization, classroom, or any group setting. It is a fundamental administrative process, serving multiple purposes such as monitoring participation, ensuring compliance with institutional policies, and supporting academic and operational planning [1]. The accuracy and efficiency of attendance tracking are crucial, as they directly impact educational outcomes, resource allocation, and institutional accountability.

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Traditional methods of taking attendance often involve manual procedures, such as roll-call or sign-in sheets, which require significant time and effort from both educators and students. Although these methods are straightforward, they are susceptible to errors and inefficiencies, including human errors, inaccuracies in recording, and potential for fraudulent activities like proxy attendance [2]. To address these limitations, several automated attendance systems have been developed [3–6], which leverage technology to streamline and enhance the accuracy of attendance recording.

One such innovation is the CCTV-based attendance system, which utilizes surveillance cameras to monitor and record classroom attendance. These systems can automatically detect and log the presence of students through visual data analysis, offering a more efficient alternative to manual methods. While CCTV-based systems improve accuracy and reduce administrative burdens, they are not without their own set of challenges.

Face recognition-based attendance systems represent a significant advancement in this domain, employing biometric technology to identify and record individuals' presence based on their facial features. Despite their potential to enhance accuracy and reduce human intervention, these systems face notable challenges. A primary concern is the occurrence of false negatives —situations where the system fails to recognize a student who is present—and false positives, where the system incorrectly identifies someone as present. These inaccuracies can undermine the reliability of the attendance records and affect the overall effectiveness of the system. Addressing these challenges is essential to improving the robustness of face recognition technologies in educational settings and ensuring their successful implementation.

This paper presents our efforts to automate student attendance at our university using facial recognition technology applied to CCTV footage installed in the classroom (top of the wall above the blackboard which the student faces). The paper is organized as follows: Section 2 provides a concise overview of pertinent research on automated facial recognition. Section 3 presents the approach and the dataset utilized. The implementation and the resulting outcomes are presented in section 4, while section 5 provides the conclusions.

2. Literature survey

The field of face recognition began to gain significant attention with the pioneering work of P. Viola and M. Jones in their 2001 paper, "Rapid object detection using a boosted cascade of simple features."[7]. This seminal work introduced foundational concepts and techniques in face recognition, setting the stage for a range of applications. The paper utilized early computer vision methods to analyze and identify facial features, marking the beginning of a new domain in computer science and artificial intelligence. The introduction of automated face recognition technology opened numerous avenues for application, extending beyond academic research into industrial and security domains [8]. In industrial contexts, face recognition has been integrated into access control systems and customer service platforms, enhancing both security and user experience. In the realm of security, face recognition has become a critical component of surveillance systems, enabling real-time identification, and tracking of individuals, thus contributing to crime prevention and public safety [9].

The application of face recognition technology to attendance systems was first explored by [10], which demonstrated the feasibility of using facial biometrics for automated attendance marking in educational settings. Their work showcased the potential of face recognition to address common issues associated with traditional attendance methods, such as manual errors and time consumption. This research laid the groundwork for the development of face recognition-based attendance systems, leading to significant advancements in both technology and application.

Historical development in face recognition-based attendance systems has seen considerable progress since these initial studies. Early systems were limited by the technological constraints of their time, including lower image resolution and less sophisticated algorithms. However, advancements in machine learning and computer vision have led to substantial improvements [11]. Current systems leverage deep learning techniques and high-resolution cameras to achieve impressive performance benchmarks. The state-of-the-art systems now report accuracy rates of approximately 96%, a significant enhancement over earlier technologies. These advancements reflect improvements in algorithmic accuracy, better handling of variations in facial appearance, and improved system robustness in diverse environments [12].

In this paper, we utilize the current benchmark technology to assess its performance in a real-world educational setting. By applying state-of-the-art face recognition systems and comparing our results with those obtained from controlled datasets, we aim to evaluate the effectiveness and reliability of these systems outside of laboratory conditions. This comparison will provide insights into the practical challenges and performance of face recognition-based attendance systems in actual classroom environments, contributing to a deeper understanding of their capabilities and limitations.

3. Data and Methods

It is suggested to use a machine learning-based face recognition system to improve the precision and effectiveness of video surveillance along with addressing issues such changing lighting and occlusions and shows notable gains over conventional techniques. [13]. Our work focuses on using fine-tuning the pre-trained benchmark deep learning (DL) models to enhance the attendance system employing face recognition technology within our institution.

The architecture diagram of our implementation is shown in Fig. 1. First we collect data of students as shown in Fig. 2 which will be used to identify an individual from a class. This dataset is then preprocessed by employing techniques such as image augmentation, enhancement, correcting illumination. This dataset will now be used as a known identity to mark attendance. Then on the application part, first frames are extracted from CCTV footage of a particular class whose attendance is to be recorded in which YoLo is used for face detection from the frames extracted through CCTV footages, then these cropped faces are preprocessed using the similar techniques as of the known individual database like image augmentation (to have multiple samples of same individual for better recognition accuracy) then we have applied image enhancement, and corrected for perspective projection. Then feature extraction is done on both the known individual dataset and the unknown one which is further used to match features using the cosine similarity method and identify a student.

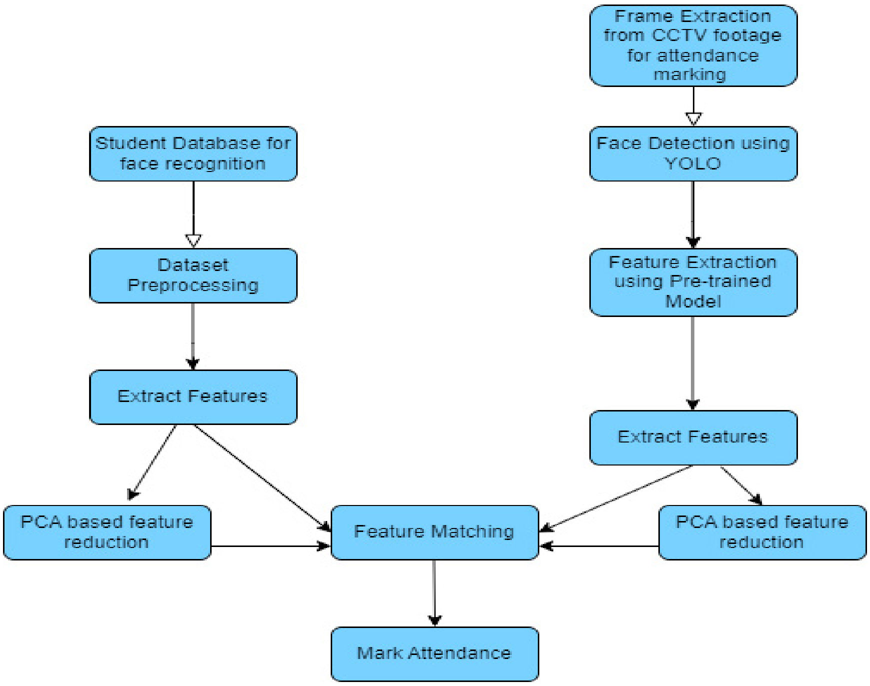

Fig. 1 Proposed Methodology

The process begins with the collection of high-quality images of individuals from a specific batch, ensuring that the dataset captures a diverse range of conditions that commonly occur in real-world settings. Data collection was conducted with the informed consent of all individuals involved, ensuring adherence to ethical guidelines and privacy standards. These conditions include various facial expressions, poses, and lighting environments, which are critical for training robust face recognition models that can perform reliably under different scenarios as shown in Fig. 2.

Fig. 2 Example of image of an individual with different pose and position

The collected dataset comprises images labelled with the identities of the individuals depicted. This labelling process is essential for supervised learning, as it provides the ground truth needed for training and validating the models. The images are meticulously categorized to reflect the diverse conditions under which they were taken, ensuring that the dataset is representative of actual classroom environments.

Prior to training, the dataset undergoes preprocessing to improve data quality and consistency. This preprocessing includes several key techniques: Image Enhancement: This involves adjusting brightness, contrast, and sharpness to improve the clarity of the images and make facial features more distinguishable. Data Augmentation: Techniques such as rotation, scaling, and flipping are applied to artificially increase the diversity of the dataset, simulating different poses and angles that might be encountered in real scenarios. Noise Addition: Controlled noise is introduced to the images to make the model more resilient to variations and imperfections that can occur in real-world settings. Correcting Perspective Projection: Adjustments are made to correct distortions caused by the angle of the camera relative to the subject, ensuring that the images have a consistent perspective. Sample images of data augmentation is shown in Fig. 3.

Fig. 3 Example of Image Enhancement and Data Augmentation

We utilised 14 images for each class. Following augmentation, we generated a total of 45 images for each class. The system required 5 to 9 milliseconds to generate a single image embedding and approximately 1.5 seconds for matching. Feature matching was accomplished via transfer learning.

The authors Wagh K, Suresh KS, Sayed M, Shahane S. [14], Demonstrated the effectiveness of transfer learning in image classification by employing pre-trained models like ResNet-50 and VGG16 for feature extraction, fine-tuning them on target datasets, and achieving significant improvements in classification accuracy, especially with limited labeled data[15]. We employed the same models in our application to extract features. For the fine-tuning process, we use two established deep learning architectures: ResNet-50 [16] and VGG-16 [17, 18]. ResNet-50, known for its deep residual learning framework as shown in Fig. 5, helps in handling complex patterns and mitigating the vanishing gradient problem, which is crucial for face recognition tasks. VGG-16, with its simpler architecture as shown in Fig. 6 but effective use of convolutional layers, offers strong performance in image classification tasks. Both models are chosen for their proven capabilities and adaptability in face recognition applications. The fine-tuning involves adapting the pre-trained models to our specific dataset by adjusting the final layers to output the correct number of classes corresponding to the number of individual identities in the dataset. Illustration of transfer learning is used in This process leverages the models learned features from large-scale image datasets and refines them to improve their performance on our institution's specific data [19]. Next, the trained models are accessed. for their accuracy and reliability in identifying individuals in various classroom scenarios, allowing us to assess their practical effectiveness in real-world conditions.

Principal Component Analysis (PCA) is employed for dimensionality reduction due to its ability to capture comprehensive information while disregarding localised aspects. PCA offers a framework for efficiently and effectively condensing a complex dataset into a lower dimension, revealing underlying structures[14]. PCA is a statistical method used to reduce the number of variables in a dataset while preserving as much of the original variability as possible. It works by identifying the directions (principal components) along which the data varies the most and projecting the data onto these directions. This reduction in dimensionality simplifies the dataset by compressing the features into a smaller set of components that capture the essential patterns in the data. Applying PCA significantly speeds up the training process by reducing the computational load associated with processing high-dimensional data. Moreover, it can help in mitigating issues such as overfitting by focusing on the most informative features.

We did through experiments on face images to determine that retaining 1,953 features out of the 4096 features extracted by the deep learning models namely VGG16 and ResNet-50 preserves 95% of the critical information from the data as shown in Fig. 4.Then we choose 1953 as optimal number of features, which hinges on achieving a balance between data reduction and utility. Applying PCA significantly speeds up the identity matching process by reducing the computational load associated with processing high-dimensional data. Moreover, it can help in mitigating issues such as overfitting by focusing on the most informative features.

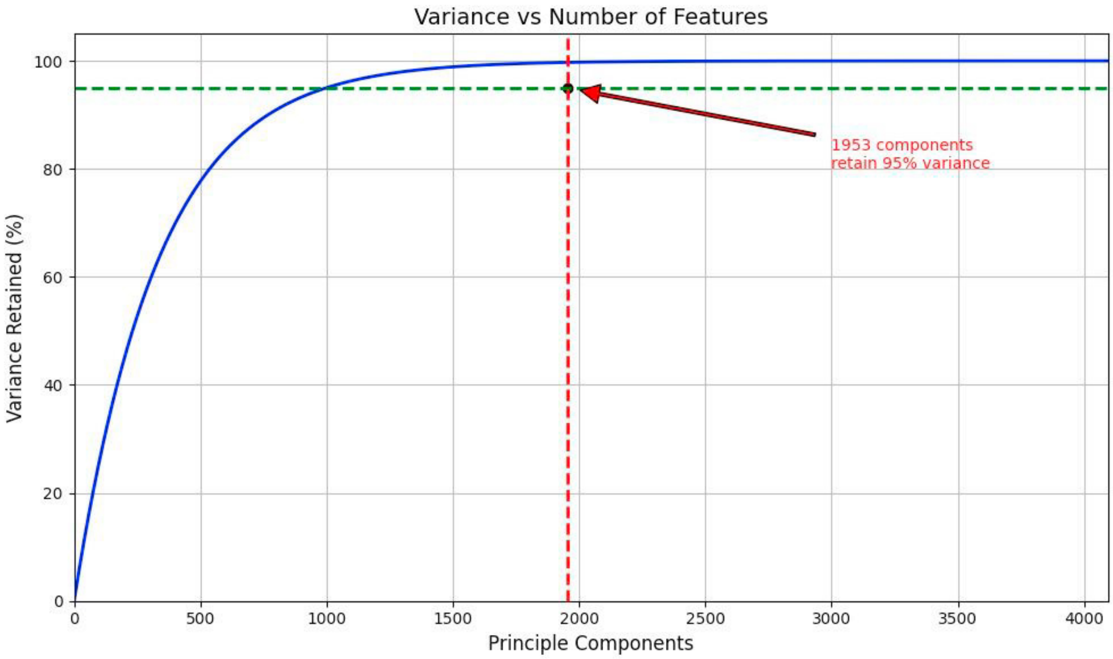

Fig. 4 Variance vs Number of features

The results of the face recognition models using PCA-reduced features are compared with those using the original, high-dimensional features. This comparison helps in evaluating the impact of dimensionality reduction on the performance and efficiency of the face recognition system. The findings from these experiments are reported in the Table 1 and Fig. 10.

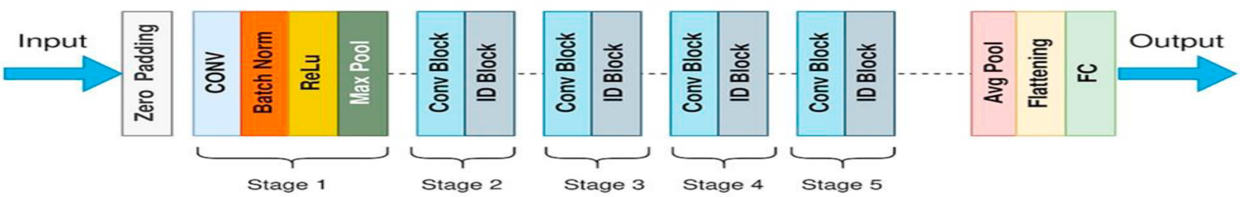

Fig. 5 RestNet50 Architecture

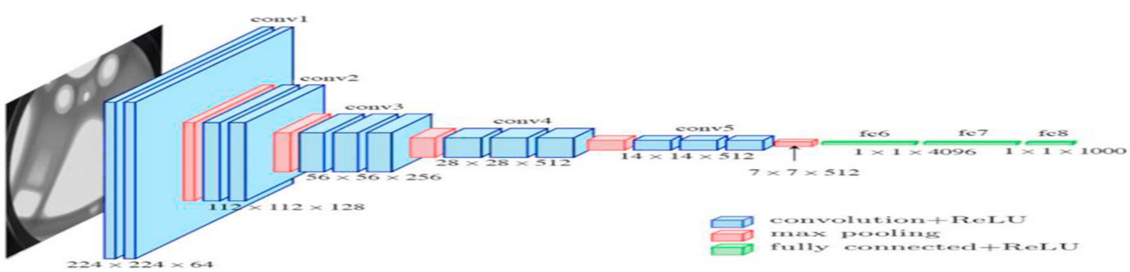

Fig. 6 VGG16 Architecture

4. Implementation and Results

It is noteworthy that Gandhi et al [20] studied the performance of transfer learning-based deep learning models for face recognition in online classes during the peak of the COVID-19 outbreak. Although the situation has since shifted to offline classes, the foundational ideas from their study remain relevant and have been adapted for our current implementation. They also highlighted that the optimal detection threshold for transfer learning-based deep learning models was 0.4, a finding that we have incorporated for our system to evaluate model performance effectively. In this study, two fine-tuned deep learning models, ResNet-50 and VGG-16, were utilized to develop an application designed for automating the attendance marking process. The application integrates with CCTV cameras installed in classrooms to capture real-time images of students. This configuration enables continuous monitoring and attendance tracking based on facial recognition. The system was further evaluated to assess its feasibility in a real-world classroom setting.

To ensure the accuracy of automatic attendance marking by our deep learning (DL) model, we tried to establish a robust criterion for verifying student presence. During a 60-minute class period, we captured CCTV frames at five equally spaced intervals. We then analyzed the distribution of recognition frequencies for each student. The student was marked present only if he/she is identified at least three times out of five. This threshold was selected to balance reliability and accuracy, ensuring that only students who are genuinely present in the class are recorded as such by the automated system.

The effectiveness of the system was measured using recall and precision rates. In the context of face recognition for attendance marking, recall (or sensitivity) refers to the ability of the system to correctly identify students who are present, minimizing the number of false negatives. A high recall rate indicates that most of the students present are accurately recognized. Precision, on the other hand, measures the accuracy of the system in identifying students as present only when they actually are, reducing how many false positives are there. High precision means that when the system identifies a student as present, it is likely to be correct.

Few examples of testing from captured CCTV frames are shown in Fig. 7 and Fig. 8.

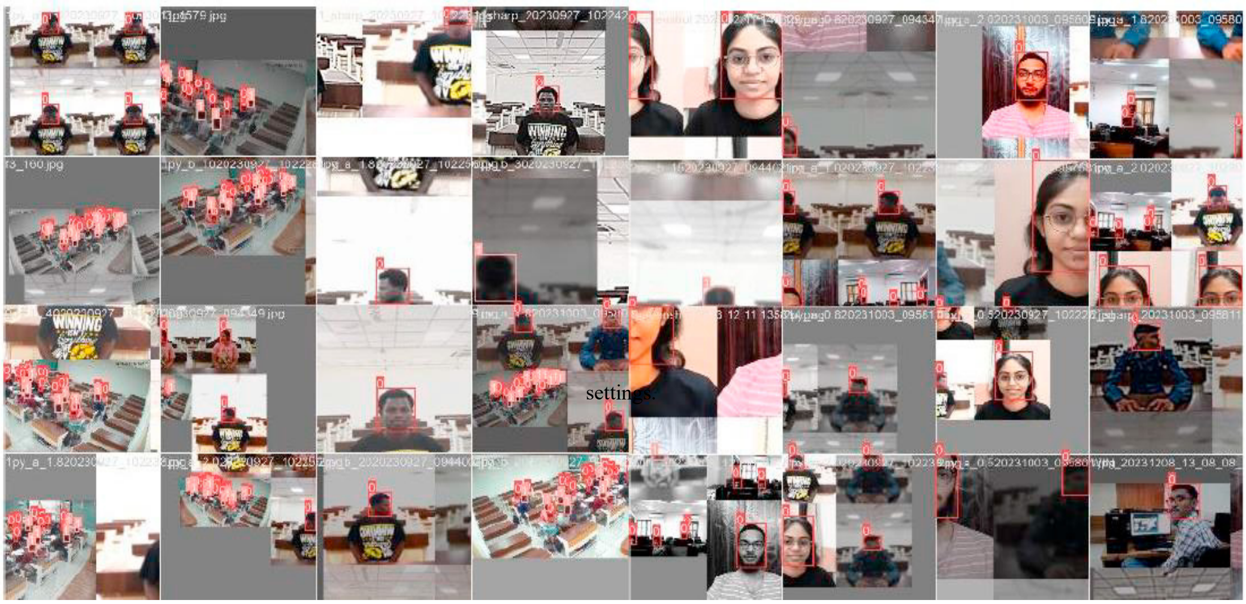

Fig. 7 Examples of detecting faces in different classroom

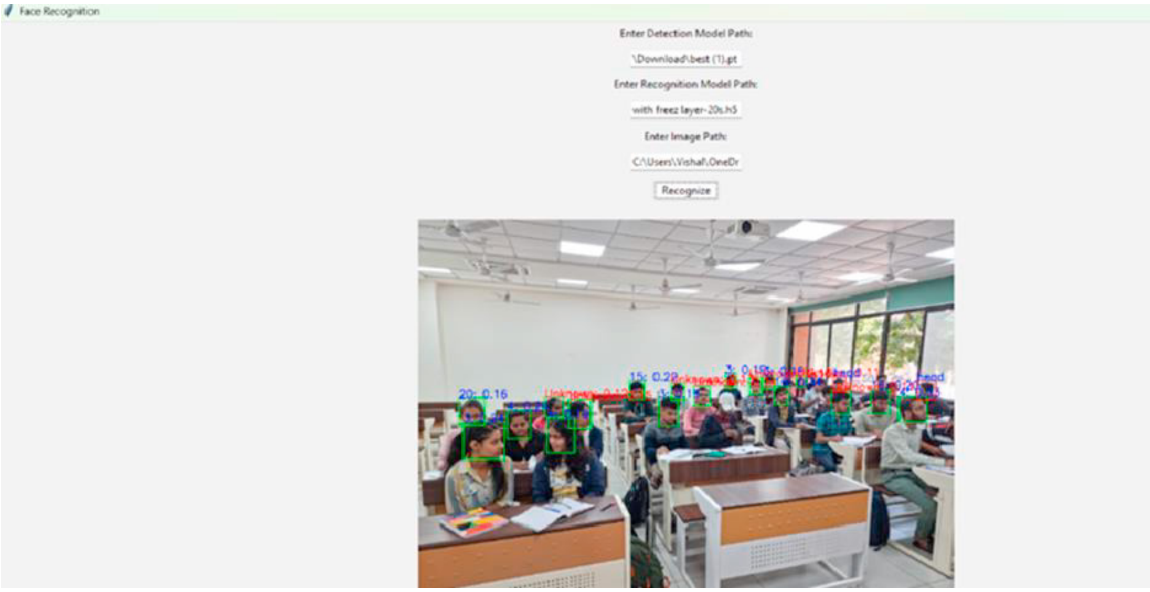

Fig. 8 Example of recognizing a student in 1 out of 5 instances

The number of times a student was marked present was then assessed for various models both with and without PCA. At the time of capturing CCTV frames, out of our database set of 41 students, 28 of them were present in the classroom. Our marking revealed that 27 out of 28 students present were successfully recognized as present by the Resnet-50 with PCA model. The outcomes for our selected models are displayed in the confusion matrix in Fig. 9.

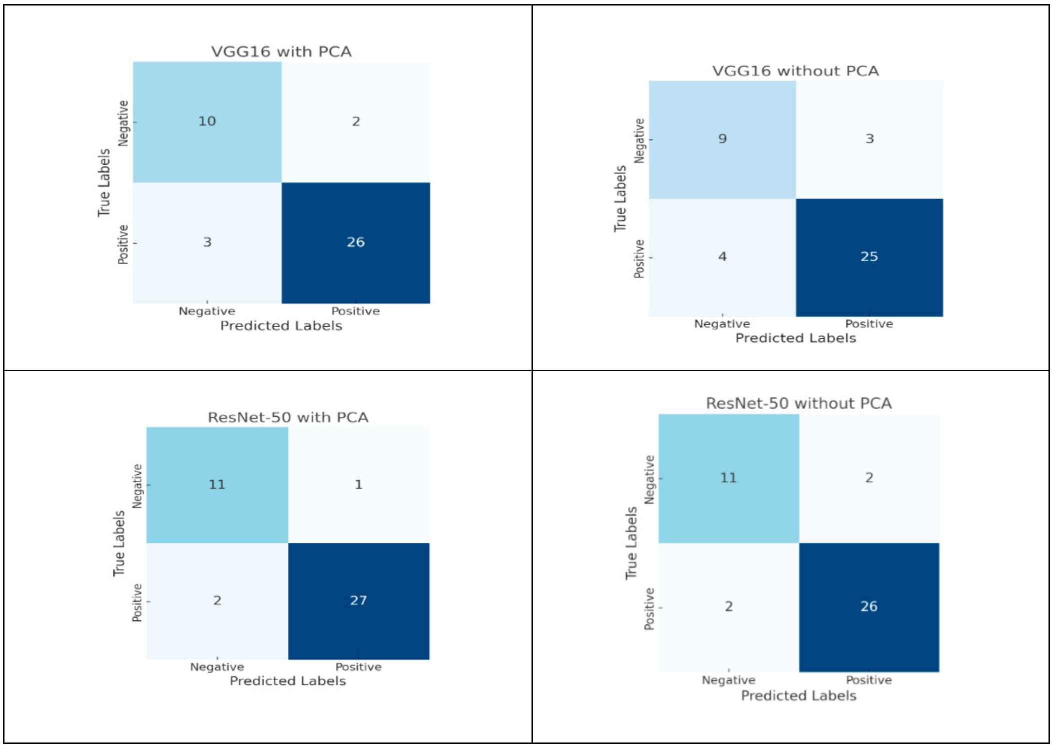

Fig. 9 Comparison of confusion matrix of two DL models with 2 different techniques

Above presented is an example of the experiment done on a single day. This experiment was repeated over the course of two weeks, the results were then analyzed and presented through performance matrices to assess the overall effectiveness of the face recognition-based attendance system. In the precision-recall analysis of our models as displayed in Table 1 and Fig. 10, ResNet-50 with PCA demonstrated superior performance, achieving the highest precision of approximately 96% and a recall of around 93%. This indicates that ResNet-50 with PCA not only accurately identifies students present in class but also effectively minimizes false negatives, ensuring reliable attendance marking.

Table 1. Results of Models testing by Accuracy, Precision, Recall and F1 Score

Model Name	Accuracy	Precision	Recall	F1 Score
VGG16 with PCA	0.878	0.929	0.897	0.913
VGG16 without PCA	0.829	0.893	0.862	0.877
ResNet-50 with PCA	0.927	0.964	0.931	0.947
ResNet-50 without PCA	0.927	0.929	0.929	0.929

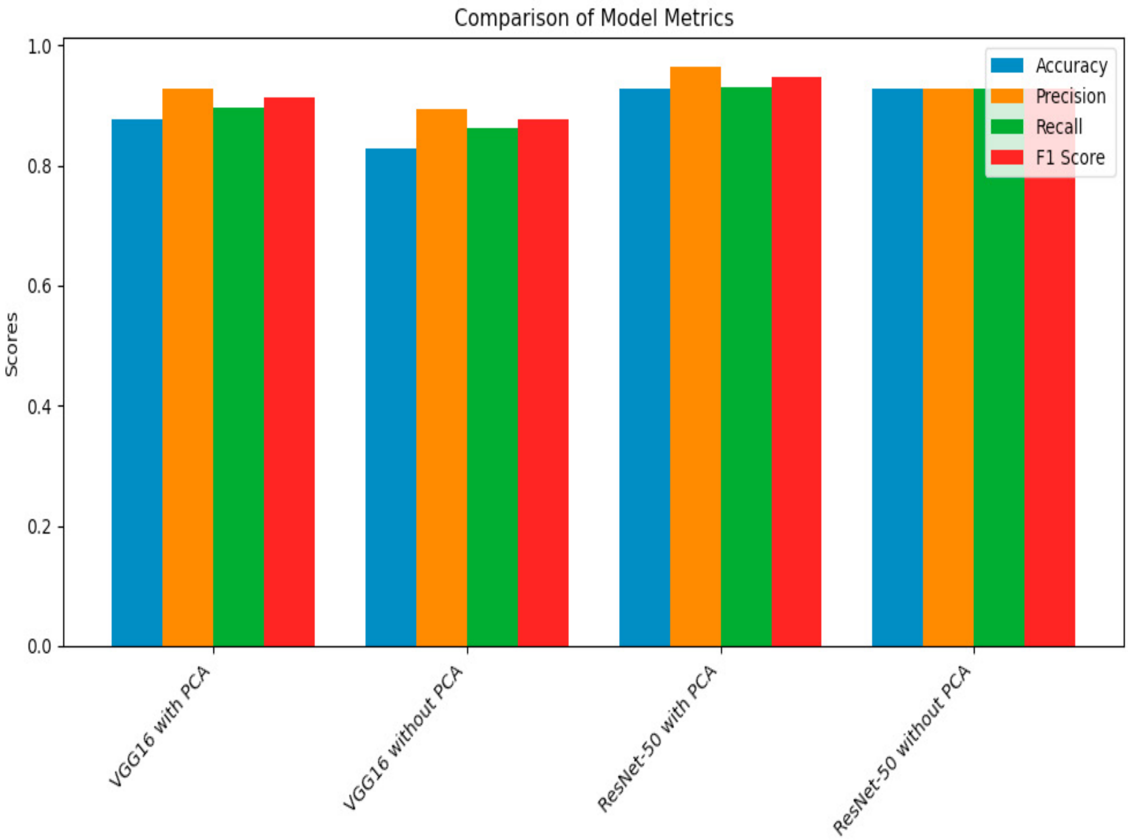

Fig. 10 Comparison of accuracy, precision, recall and F1 Score

5. Discussion and Conclusion

This research is aimed towards developing an automated attendance system that can work in real world scenario such as a classroom. The precision and recall analysis reveal that the ResNet-50 model, when combined with PCA for feature reduction, outperformed all the other techniques which we experimented with (ResNet50 and VGG16 as feature extractors and then using them for student identification with and without PCA as feature extractor) . This model demonstrated superior performance over VGG16 in identifying students accurately, reflecting both high precision and recall rates. Over the course of the experiment, ResNet-50 achieved the highest accuracy at 93%. This outstanding performance underscores the model's robustness and reliability in a practical setting, reinforcing its potential as the most effective solution among those tested.

The results of this study indicate that the identifications done using Principal Component Analysis (PCA) extracted features perform significantly better than those using all features. PCA effectively reduces the dimensionality of the data by focusing on the most important features for face recognition, which enhances the models' performance. By extracting the key features necessary for accurate identification, PCA helps in improving the overall accuracy and efficiency of the face recognition system.

The experiment conducted demonstrated that the system performs well in an actual classroom setting, accurately identifying most students at a time as shown in fig. 8. and eliminating the need of one-by-one roll call which requires a considerable amount of time. This suggests that the system is robust and effective under real-world conditions. However, achieving 100% detection accuracy remains challenging. Several factors contribute to this limitation. Firstly, the quality of the CCTV cameras used for capturing images can affect the system's performance, as lower resolution or poor-quality images can hinder accurate face recognition. Secondly, the base architecture of the models themselves does not guarantee perfect detection rates, as inherent limitations exist in current face recognition technologies.

Despite these challenges, the system's performance is promising, but there is room for improvement. Ensuring that no student is marked absent when present is crucial for the system's success. Potential improvements include upgrading to higher-quality cameras to capture clearer images, though this may involve significant costs. Another alternative is employing image enhancement techniques using generative models. These models can improve video quality through post-processing, which could enhance face recognition accuracy.

Additionally, the pre-trained models used in this study are not perfect and could benefit from further refinement. This work provides valuable insights into the factors affecting face recognition performance and highlights areas for future development. By addressing these factors and exploring new approaches, such as advanced generative models and improved camera technologies, the accuracy and reliability of face recognition-based attendance systems can be significantly enhanced for practical use in educational settings.

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